IPCA ESCOLA SUPERIOR DE TECNOLOGIA

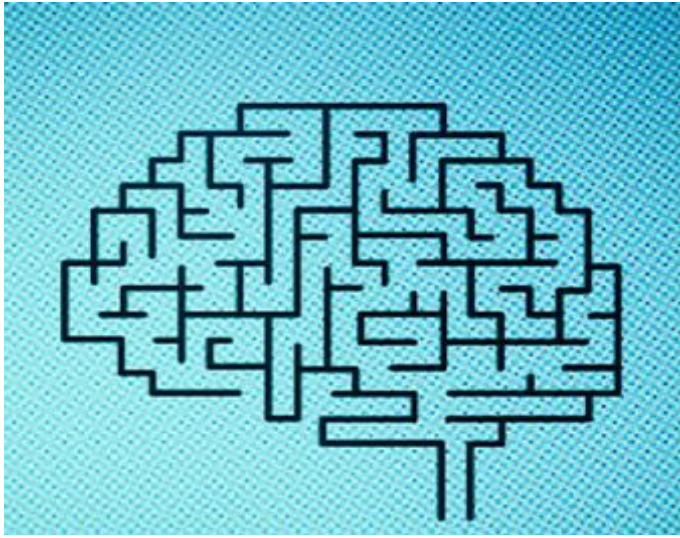
Project Guide 01

Artificial Intelligence (AI), 2026-27

Degree on Computer Systems Engineering

Introduction

- The goal of the project is to develop an intelligent agent, documenting the design and implementation steps
- Each work group needs to be made up of **4 students**
- The project must be available in a **Github repository**
  - The code and documentation must be merged into a Jupyter Notebook
  - The folder must contain all files required to run the solution
  - The folder must be shared with the lecturer
- A ZIP file of the repository must be submitted on Moodle before the deadlines
- The presentation and defense of the project will take place on a date to be scheduled later.



Collaborative Platforms

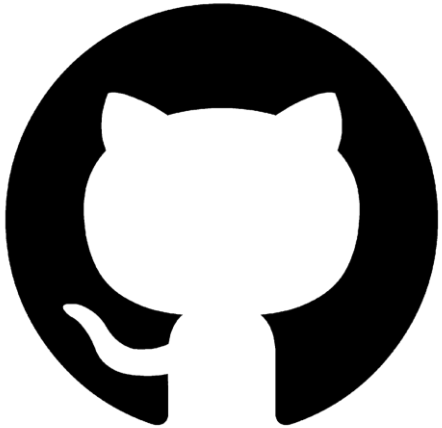
The project should be hosted in a private **GitHub** repository

- Use the nomenclature "**IA26_P01_G##**", where G## corresponds to the group
- The lecturer must be added to the repository

The project must be submitted as a ZIP file labelled **IA_P01_G##.zip**, where G## corresponds to the group

The file must contain

- code files, including a Jupyter Notebook;
- readme.txt file with the link to the repository (Github);
- other files needed to run the notebook.



Theme — Local food

Motivation

- Growing EU demand for local food, but small producers face high logistics costs and price volatility
  - Short supply chains cut transport emissions and food waste compared to long-distance distribution
  - Fruit and vegetable prices have risen sharply in the EU in recent years, squeezing both producers and consumers
- A platform connecting producers, consumers and logistics can make local distribution efficient and viable
  - The key challenge is routing — selecting efficient pickup-and-delivery routes under time-window constraints

Business model

- A digital platform connecting three types of actors
  - **Producers** — small local farms offering fruits, vegetables and other products
  - **Customers** — place orders and choose a delivery location
  - **Collection centers (warehouses)** — aggregate products before distribution, operating a fleet of vehicles
- The platform's routing agent selects the best pickup-and-delivery routes
  - Considering pickup locations (producers), drop-off locations (chosen by each customer), and the distances/times between all locations
  - Respecting the time windows in which products are available at the producer and must be delivered to the customer

Algorithms to apply

- Informed search (e.g., A*, greedy best-first)
  - Find efficient pickup-and-delivery routes using a distance/time heuristic
- Local search (e.g., hill climbing, simulated annealing)
  - Improve an initial routing solution by iteratively exploring neighboring route configurations
- Constraint Satisfaction Problems (CSPs)
  - Model time-window constraints (availability at producers, delivery deadlines at customers) and vehicle capacity as constraints to satisfy

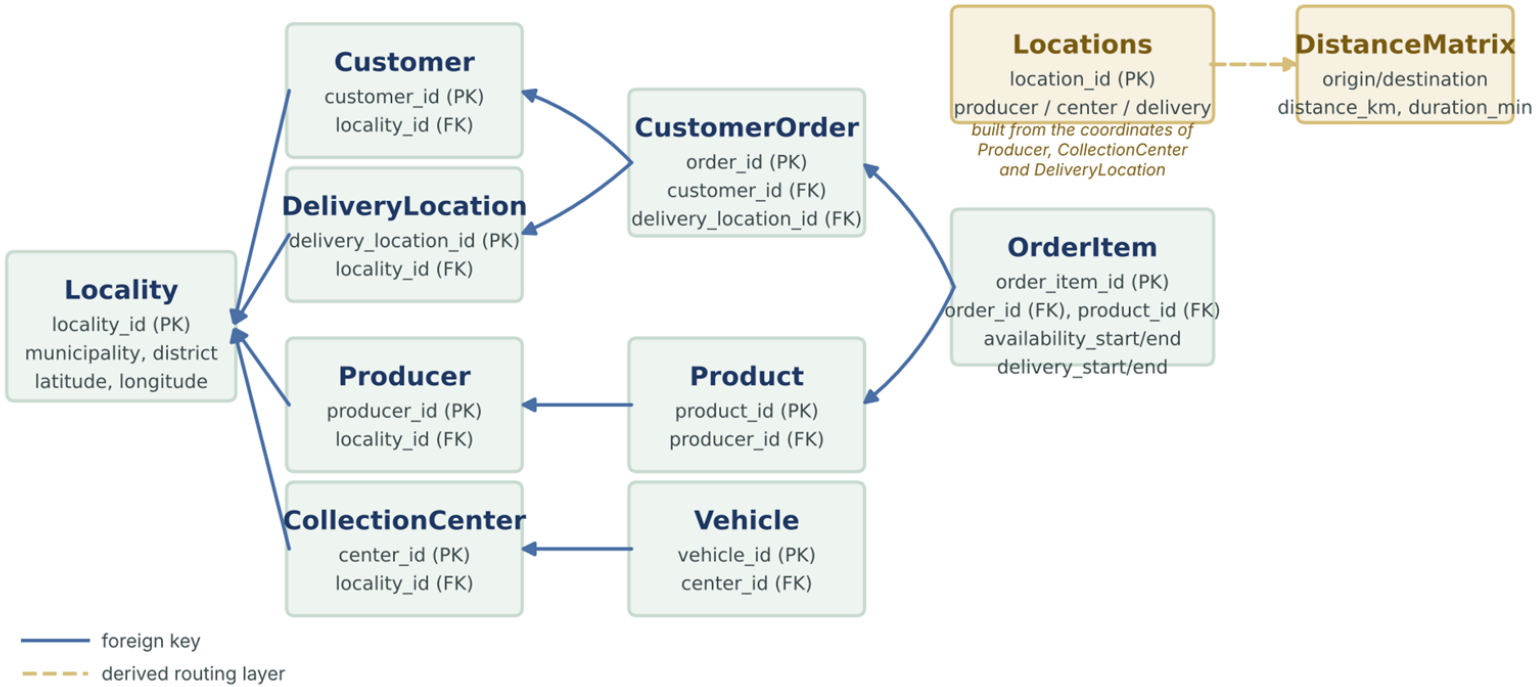
Dataset — Overview

- A synthetic dataset supports the project, generated as a set of CSV files (no database engine required)
  - Covers the Braga district and includes Póvoa de Varzim
- Captures everything the routing agent needs
  - Producers, collection centers (warehouses), vehicles, customers and delivery locations
  - Customer orders with pickup and delivery time windows
  - A distance-and-time table between every pair of routing stops

Dataset — Entities and Relationships

- **Locality** — shared geocoded place (parish, municipality, district, coordinates) used by producers, centers, delivery locations and customers
- **Producer** — small local producer, located at a Locality, offering one or more Products
- **CollectionCenter** — warehouse where goods are aggregated and home base for the Vehicles
- **Customer** — places CustomerOrders and chooses a DeliveryLocation for each one
- **CustomerOrder / OrderItem** — one order per customer; each item has an availability window at the producer and a delivery window at the delivery location

Dataset — Schema Diagram



Dataset — Orders and Time Windows

- Each customer order is linked to exactly one delivery location, chosen by the customer
- Each order line (OrderItem) carries two time windows
  - **availability_start / availability_end** — when the product can be collected at the producer
  - **delivery_start / delivery_end** — when it must arrive at the delivery location
- These windows are exactly the constraints the routing agent must satisfy
  - Pickups cannot happen before a product is available; deliveries cannot happen after the delivery window closes

Dataset — Distance and Travel-Time Matrix

- ***distance_matrix*** gives distance and travel time between every ordered pair of routing stops (producers, collection centers, delivery locations)
  - **distance_km** — great-circle distance scaled by a route factor (default 1.3) to approximate real road distance
  - **duration_min** — distance converted to time using an average road speed (default 45 km/h)
- A synthetic approximation, but deterministic and symmetric
  - Enough to exercise informed search, local search and CSP time-window constraints without needing a live maps API

Jupyter Notebook

Tools

- The project must be implemented in **Python** using a **Jupyter Notebook**
- Two datasets will be available in Moodle
  - a small, simpler dataset
  - a larger, more complex dataset
- The notebook must be well structured
  - Each step in a distinct block
  - Text blocks documenting the reasoning and decisions
  - Comments in the code blocks

Notebook structure

- Introduction — Preliminary delivery
  - Establish here the context and the purpose of project
  - Identify the teammates: student name and number

- Exploratory Data Analysis (EDA) — Load CSVs and explore data
  - Validate integrity and time windows, measure weights vs. capacities, and map stops.
  - Deliverable: 01_eda_tiny.ipynb notebook.

- Modeling — Define Problem, Request, Vehicle, and State classes;
  - load instances from CSVs; implement cost function and route feasibility checker.
  - Deliverable: model.py module + tests with hand-written routes.

Notebook structure (2)

- Problem Solving
  - Informed search — Greedy and A* search with 2 heuristics; count expanded nodes.
    - Deliverable: Routes for the 2 tiny vehicles.
  - Local search — Hill-Climbing on top of the Greedy solution
    - Deliverable: Cost vs. iteration curves.
  - CSP (Constraint Satisfaction Problem) — Formulate the problem
    - Present the variables, their domain and the constraints definition
    - Highlight the heuristics applied to implement the soft constraints
    - Deliverable: Feasible assignment and number of backtracks.

Notebook structure (3)

- Result Analysis
  - Provide the best solutions you can find
  - Perform a critical analysis of the results
    - Provide execution times
    - Compare the results from different search algorithms
    - identify some future improvements to the agent

- Conclusion
  - Draw up a conclusion about the outcomes, the development process and the tools used.
  - The structure of the notebook can be adapted according to each project characteristics.

Thank you!